

SAMKELO SITHEMBISO MNGOMA

SOFTWARE & DATA ENGINEERING | CLOUD DELIVERY | TECHNICAL OPERATIONS

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Portfolio: mngomazar.github.io/samkelo-mngoma-portfolio

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PROFESSIONAL PROFILE

Hands-on software and data practitioner with experience in database administration, analytics, technical operations and cloud delivery. Co-founder and COO of SAICTS, combining stakeholder coordination with implementation across mobile applications, learning platforms and data systems. Seeking software, data, cloud delivery or technical operations opportunities.

CORE SKILLS

Software: Python, TypeScript, JavaScript, React Native, FastAPI, Node.js, REST APIs.

Data: SQL, PostgreSQL, Power BI, data cleaning, transformation, modeling and reporting.

Delivery: Git/GitHub, Docker, GitHub Actions, Oracle Cloud, Linux, Expo/EAS, regression and integration testing.

Operations: Requirements, documentation, stakeholder communication, role-based access, secret hygiene and release coordination.

PROFESSIONAL EXPERIENCE

Co-founder & Chief Operating Officer

SAICTS | Mar 2024 - present

- Coordinate project delivery, operational processes, resources, risks and stakeholder communication.
- Contribute hands-on software, API, data and cloud implementation for learning and marketplace platforms.
- Maintain deployment documentation, test evidence and release checks across GitHub, Oracle and Expo delivery workflows.

Data Specialist (contract)

Lindbong Development | Jan 2023 - Mar 2024

- Cleaned, transformed and analyzed data; prepared visualizations and reports to support business decisions.
- Translated analysis into stakeholder-facing insights and coordinated technical requirements.

Database Administrator (contract)

Colman Protection | Aug - Dec 2023

- Worked on database setup, cloud configuration, access control, backup/recovery and integration planning.

Customer Retention & Data Operations

AA Wealth (insurance) | 2020 - 2023

- Supported customer retention, data processing and cleaning, financial-data review and business communication.

SELECTED PROJECTS & DEVELOPMENT

SELECTED ENGINEERING & ANALYTICS WORK

Madabukela Academy - integrated learning platform

SAICTS technical delivery | 2026 | live public platform

- Worked across onboarding, assessments, learner evidence portfolios and placement/in-service structures.
- Implemented offline queue handling, immediate quiz feedback, context-aware tutor routing and tablet-oriented interfaces.
- Used Node.js, PostgreSQL, Docker and Oracle Cloud; verified public desktop/tablet flows.
madabukela-academy.co.za

Papzi - creator marketplace

SAICTS contracted engineering | 2026 | deployed TestFlight beta

- Worked across Expo/React Native, FastAPI/PostgreSQL, role access, scheduled bookings, messaging, road routing and release configuration.
- Delivered Oracle Docker/GHCR backend and TestFlight beta; CI verified 300 concurrent read-test sessions and 3,000 requests without errors.
- Restricted beta, not a completed public financial marketplace; live payment and device acceptance remain ongoing.

Commercial performance reporting

Independent Power BI assignment | 2026

- Built native multi-page sales reporting with executive, regional, sales-manager, brand-trend and customer views, interactive filtering and recommendations. Private datasets are not public; no AB InBev/SAB employment claim.

ESP32 telemetry & Phepha safety intelligence

Engineering prototypes | 2026

- ESP32/MPU6050 firmware: sensor fusion, receiver decoding, serial diagnostics and arming safety gates; hardware acceptance remains incomplete.
- Phepha: Express/WebSocket API aggregation, MapLibre layers, TTL caching and source-attributed public-data integration.

EDUCATION & PROFESSIONAL DEVELOPMENT

NCV Finance, Economics & Accounting, Levels 2-4

Esayidi TVET College | 2020-2022

Online courses and certificates: IBM Applied Data Science; Google Data Analytics; IBM data engineering, DevOps, CI/CD, application security and cloud computing; Microsoft/freeCodeCamp Foundational C#; Alison database systems, computer networking and cybersecurity (online CPD). Certificate copies available on request.

Virtual job simulations: KPMG Data Analytics and JPMorgan Chase analyst/private-banking exercises via Forage (2023). These are training simulations, not employment or internships.

LEADERSHIP

Esayidi TVET SRC President (2021/2022) and Secretary General (2022/2023): student representation, governance, administration and stakeholder coordination.